

NAHEED ANJUM ARAFAT

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RESEARCH INTERESTS & EXPERTISE

I design adversarially robust, scalable, explainable, and certifiable AI systems for analyzing complex data arising from interactions among entities. In particular, I study learning systems involving **graphs/hypergraphs** and their robustness under adversarial attacks.

EMPLOYMENT HISTORY

Post-doctoral Research Fellow

College of Engineering and Architecture, Howard University

Washington D.C., USA

July 2025 - Now

Post-doctoral Research Fellow

Rolls-Royce@NTU Corporate Lab, NTU

Singapore

July 2021 - 2024

Post-doctoral Research Fellow

Department of Information Systems & Analytics, NUS

Singapore

July 2020 - July 2021

Research Assistant

Department of Computer Science, NUS

Singapore

Nov 2019 - June 2020

Full-time Lecturer

Department of Computer Science, United International University

Dhaka, Bangladesh

Jun 2014 - Jul 2015

RESEARCH EXPERIENCE

Post-doctoral Research Fellow

PI: Professor Danda B. Rawat

College of Engineering and Architecture, Howard University

March 2025 - Now

- **Projects:** Develop, design, and evaluate secure, trustworthy, robust, fair, explainable AI systems and ML algorithms.

Post-doctoral Research Fellow

PI: Associate Professor Adams Wai Kin Kong

Rolls-Royce@NTU Corporate Lab, NTU

July 2021 - 2024

- **Projects:** (1) Super-resolution/-fidelity of CFD simulations. (2) Accelerating unsteady flow simulation with physics-based AI.

EDUCATION

National University of Singapore (NUS)

PhD

Singapore

Aug 2015 - Nov 2020

- **Thesis:** Analysis and generation of data with topology from combinatorial representations

Bangladesh University of Engineering and Technology (BUET)

Bachelors in Computer Science & Engineering

Dhaka, Bangladesh

Mar 2009 - Jul 2014

- **Thesis:** A Fuzzy Dominance Based Evolutionary Algorithm for solving many-objective optimisation problems.

RESEARCH PUBLICATIONS

• Conference Proceedings

1. **Naheed Anjum Arafat** and Danda B. Rawat. Supervised incidence sparsification of hypergraphs under budget constraints. In *Learning on Graph Conference (LoG)*, 2026
2. **Naheed Anjum Arafat**, Debabrota Basu, Yulia Gel, and Danda B. Rawat. Adversarial attacks and robust training for hypergraph neural networks. In *Forty-third International Conference on Machine Learning (ICML)*, 2026 [pdf]
3. Siddhartha Shankar Das, **Naheed Anjum Arafat**, Muftiqur Rahman, S M Ferdous, Alex Pothan, Mahantesh Halappanavar, and Danda B. Rawat. SGS-GNN: A supervised graph sparsifier for graph neural networks. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, 2026 [pdf]
4. Bishwamittra Ghosh, Sarah Hasan, **Naheed Anjum Arafat**, and Arijit Khan. Consistency of language models for logic facts checking on knowledge graphs. In *International Conference on Learning Representations (ICLR)*, 2025 [pdf]
5. **Naheed Anjum Arafat**, Debabrota Basu, Yulia Gel, and Yuzhou Chen. When witnesses defend: A witness graph topological layer for adversarial graph learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025 [pdf]
6. **Naheed Anjum Arafat**, Ehsan Bonabi Mobaraki, Arijit Khan, Yllka Velaj, and Francesco Bonchi. Estimate and reduce uncertainty in uncertain graphs. In *IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2024 [pdf]

7. *Loh Sher En Jessica, ***Naheed Anjum Arafat**, Wei Xian Lim, Wai Lee Chan, and Adams Wai Kin Kong. Finite volume features, global geometry representations, and residual training for deep learning-based cfd simulation. ICML ([spotlight](#)), 2024. * = equal contrib [[pdf](#)]
8. Wei Xian Lim, **Naheed Anjum Arafat**, Wai Lee Chan, and Adams Wai Kin Kong. Multi-order loss functions for accelerating unsteady flow simulations with physics-based ai. In *IEEE Conference on Artificial Intelligence (CAI)*, 2024 [[pdf](#)]
9. **Naheed Anjum Arafat**, Arijit Khan, Arpit Kumar Rai, and Bishwamittra Ghosh. Neighborhood-based hypergraph core decomposition. volume 16. VLDB Endowment, May 2023 [[pdf](#)]
10. **Naheed Anjum Arafat**, Debabrota Basu, Laurent Decreusefond, and Stéphane Bressan. Construction and random generation of hypergraphs with prescribed degree and dimension sequences. In *Database and Expert Systems Applications*, pages 130–145. Springer International Publishing, 2020 [[pdf](#)]
11. **Naheed Anjum Arafat**, Debabrota Basu, and Stéphane Bressan. Topological data analysis with ϵ -net induced lazy witness complex. In *Database and Expert Systems Applications*, pages 376–392. Springer International Publishing, 2019 [[pdf](#)]
12. **Naheed Anjum Arafat** and Stéphane Bressan. Hypergraph drawing by force-directed placement. In *International Conference on Database and Expert Systems Applications*, pages 387–394. Springer International Publishing, 2017 [[pdf](#)]

• Journals

13. Siddhartha Shankar Das, Md Monirul Islam, and **Naheed Anjum Arafat**. Evolutionary algorithm using adaptive fuzzy dominance and reference point for many-objective optimization. *Swarm and evolutionary computation*, 44:1092–1107, 2019 [[pdf](#)]

• Patents

14. Jessica Sher En Loh, **Naheed Anjum Arafat**, Wai Kin Adams Kong, Wai Lee Chan, and Bryce D. Conduit. Fluid flow simulation, Oct 2024. Granted; Filed: 14 August, 2023; Granted: 8 October, 2024
15. Jessica Sher En Loh, **Naheed Anjum Arafat**, Wai Kin Adams Kong, Wai Lee Chan, Bryce D. Conduit, Wei Xian Lim, and Thant Zin Oo. Fluid Flow Simulation, Feb 2025. Pending; Priority: Aug. 14, 2023; Filed: July 30, 2024

• Workshops/Posters

16. **Naheed Anjum Arafat**, Debabrota Basu, and Stéphane Bressan. ϵ -net induced lazy witness complexes on graphs. *Workshop on Applications of Topological Data Analysis (ATDA)*, held in conjunction with ECML-PKDD, 2019 [[pdf](#)]

• Preprints/Under-review

17. **Naheed Anjum Arafat** and Danda B Rawat. Certifiable robustness of hypergraph neural networks. 2026. (under review)
18. Mikel K. Ngueajio, **Naheed Anjum Arafat**, and Danda B. Rawat. Evaluating hatefulness in fake news through llm and human alignment. 2026. (under review)
19. **Naheed Anjum Arafat** and Danda B. Rawat. Auditing explanation preservation in sparsified gnn pipelines. 2026. (under review)

TEACHING EXPERIENCE

As Graduate Assistant

NUS

Jan 2016 - 2019

I conducted weekly tutorials and held one-on-one consultation sessions for the following courses:

- **Big Data Techniques and Technologies (BT4221)**: Department Of Information Systems and Analytics, School of Computing - Semester 1, 2019.
- **Data Management and Warehousing (BT5110)**: Master Of Science Business Analytics Programme - Semester 1 of 2019, 2018 and 2017.
- **Information Visualisation (CS5346)**: Department of Computer Science, School of Computing - Semester 2, 2018.
- **Discrete Structures (CS1231)**: Department of Computer Science, School of Computing - Semester 1, 2016.
- **Programming Methodology (CS1010E)**: Department of Computer Science, School of Computing - Semester 2 of 2017 and 2016.

As Full-time Lecturer

United International University

I designed course curriculum, prepared lecture notes, assignments, and exam questions, delivered lectures, conducted labs, and evaluated assignments and exam scripts. I was also involved in various administrative duties. The courses include:

- **Advanced Programming Languages**: Summer 2014, Fall 2014, Spring 2015, Summer 2015 trimesters.
- **Data Structures**: Fall 2014, Summer 2015 trimesters.
- **Discrete Mathematics**: Summer 2014 trimester.
- **Structured Programming Language**: Spring 2015 trimester.
- **Electric Circuits**: Fall 2014, Spring 2015, Summer 2015 trimesters.

MENTORING EXPERIENCE

I am fortunate to work with and train the following students in academic research:

- **Mikel K. Ngueajio:** PhD student at Howard University, USA. (2025-Present)
- **Siddhartha Shankar Das:** PhD student at Purdue University, USA. (2024-2025)
- **Sarah Hasan:** PhD student at Aalborg University, Denmark. (2023-2024)
- **Ehsan Bonabi Mobaraki:** PhD student at Aalborg University, Denmark. (2022-2023)
- **Loh Sher En Jessica:** PhD student at Nanyang Technological University. (2021-2023)
- **Bishwamittra Ghosh:** PhD student at National University of Singapore. (2021-2022)
- **Arpit Kumar Rai:** Undergraduate student at IIT Kanpur (2021-2022)
- **Debabrata Mahapatra:** PhD student at National University of Singapore (2019)

RESEARCH MANAGEMENT EXPERIENCE

- **Research Equipment Purchasing**
 - I led the acquisition of a high-performance GPU server for experimenting with deep neural networks.
 - * I managed the entire procurement process, from preparing requirement specifications for Invitation-To-Quote (ITQ) and setting clear evaluation criteria to evaluating potential vendors and overseeing the installation, testing & commissioning of the server.
 - * I collaborated with the IT department to ensure seamless integration of the new equipment into ongoing research projects, significantly enhancing the computational capabilities of RR@NTU Corp Lab.
- **Grant Writing**
 - AI Singapore Research Grant.
 - * I collaborated with my PI (at NTU) on developing the grant proposal by proposing ideas for potential work packages and drafted sections of the grant application, such as discussing state-of-the-art and their limitations.

TALKS

- **Guest Lectures:**
 - * **Big Data Techniques and Technologies module (NUS):** I delivered two guest lectures on 'Practical data analysis using Amazon EMR and Sagemaker services' during Semester 1, 2020 offering of the undergraduate module BT4221.
 - * **Data Management and Warehousing module (NUS):** I delivered a hands-on lecture on 'developing retail-sales data mart using Pentaho' during Semester 1, 2018 offering of the masters module BT5111.
 - * **Information visualisation module (NUS):** I delivered a guest lecture on 'Graph and hypergraph data visualisation tools' during Semester 2, 2018 offering of the master's module CS5346.
- **Seminar Talks:**
 - * **Talk at Grand Valley State University, MI (2025):** Graph Data Mining and Learning: Modern Applications, Challenges and Paradigms
 - * **Moscow-Beijing Topology seminar (2022):** I presented my work on Topological data analysis.
 - * **DEXA (2020):** I presented my paper: *Construction and random generation of hypergraphs with prescribed degree and dimension sequences*
 - * **DEXA (2019):** I presented my paper: *Topological data analysis with ϵ -net induced lazy witness complex*.
 - * **E2S2-CREATE seminar (2019):** I gave a talk on *Topological Data Analysis* at E2S2-CREATE Seminar, Feb 2018 attended by 20+ researchers from E2S2 project funded by NRE

ACADEMIC SERVICES

- **Editorial Board Member:** Knowledge Engineering Review (KER) (2025-present)
- **PC Member:** AAAI (2026-27), ICML (2025), ICLR (2025-26), NeurIPS (2024-2026), AI for Social Impact Track @ AAAI (2026-27), WSDM (2026), Australasian Database Conference (2025), MLG workshop @ECML-PKDD (2025), LLM+G Workshop@VLDB (2025), CODS-COMAD (2024-2025), ICDE (2025, 2018), LoG (2024), CIKM (2024), DASFAA (2020), DAWAK (2020), VLDB (2017), DEXA (2017), SKIMA (2014).
- **Journal Referee:** IEEE Transactions on Information Forensics & Security (TIFS), IEEE Transactions on Artificial Intelligence (TAI), Journal of Applied and Computational Topology (JACT), IEEE Transactions on Knowledge and Data Engineering (TKDE).
- **Session Chair:** VLDB 2023 ([Learning, Recommendations, Social Networks](#))
- **Co-organiser:** Co-organised *8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA)*, 2014
- **Session Chair:** Chaired two sessions at *NUS SoC-Telecom ParisTech joint workshop on Data Sciences and Artificial Intelligence*, April 2018 attended by 20+ scholars from Telecom ParisTech and NUS.
- **Session Chair:** Chaired sessions at *NUS School of Computing Research Workshop*, March 2018 attended by 50+ scholars across Asia.

AWARDS & ACHIEVEMENTS

- **Distinguished PC member award:** [CODS-COMAD 2024](#)
- **NUS Research Scholarship** for PhD students (2015-2019).
- **Teaching Assistant Training Certificate** issued by the Centre for Development of Teaching & Learning (CDTL), NUS on applying 'Collaborative Learning' approaches.
- **Distinguished Poster Award** at Undergraduate Research Poster Presentation Workshop 2014, Department of Computer Science and Engineering, BUET for presenting my undergraduate Capstone project
- **Inter-University Hardware Project Show Champion** award for showcasing the project titled 'Cell phone operated mini-car with surveillance' at CSE festival 2013, BUET.
- **Imdad-Sitara Khan Scholarship (Bangladesh)** for high-school studies (2006-2008) and undergraduate studies (2009-2013).
- **Education Board scholarships** for excellence in Secondary and Higher Secondary exams, Jashore Board, Bangladesh.

REFERENCES

Available upon request.